Weekly Report

Simplified Version, Week of June 30, 2026

1 Mechanisms

This week focuses on two datapath batching mechanisms in the real simulator.

Mechanism 1 targets the cache/DRAM path. It keeps two simple optimizations:

- **L2 directory lookup batching:** requests that look up the same L2 directory set can reuse part of the set/way lookup work.
- **DRAM access-unit coalescing:** two adjacent 64B cache lines that fall into the same 128B DRAM access unit can be merged before DRAM timing.

Mechanism 2 targets remote access. It adds a requester-side RDMA batch queue so nearby remote requests can be grouped before entering the wafer network path.

The experiment runner now supports four modes: baseline, M1, M2, and M1+M2. This allows a direct ablation study under the same benchmark, sampling, and maximum-workgroup settings.

2 End-to-End Results

The main speedup results come from the June 29 ablation run. For workloads where baseline, M1, M2, and M1+M2 all produced comparable full results, the current end-to-end result is small: M1+M2 improves geometric mean performance by only 1.004×. M2 helps FloydWarshall, but PageRank slows down under M1+M2, and most other comparable workloads are close to neutral.

Figure 1 shows the broader available result set. Some large bars come from runs that did not reach the same max-workgroup condition, so the conservative conclusion should be based on the comparable full-result subset.

3 Why Speedup Is Small

M1 and M2 should be viewed as data-access datapath optimizations rather than guaranteed end-to-end application optimizations. As shown in Figure 2, M1+M2 reduces the average cycles per traced memory path for many workloads, which shows that the optimized datapath is active and often faster. However, the remaining latency can still be large: remote-heavy workloads such as matrix transpose, PageRank, Im2Col, and SpMV still spend hundreds of cycles per access in the network, while KMeans is nearly unchanged. Therefore, application speedup is small when the optimized portion is not the dominant remaining bottleneck, or when the improved memory path does not cover enough of the full execution time.

4 Remote-Network Motivation

I added a remote-only streaming memory-path trace to record completed remote L1V memory paths without keeping all raw records in RAM. The baseline results show that network/RDMA is the dominant part of the remote memory path in these workloads. Net. share is the fraction of remote memory-path latency spent in network/RDMA, which ranges from 80.8% to 95.6% in the table. GPU max/min is the ratio between the largest and smallest requester-GPU average network latency, showing that different GPUs can experience very different network costs, up to 8.80× for KMeans.

Table 1: Remote-only trace highlights.

Workload	Net. share (%)	GPU max/min
SpMV	92.7	4.20×
MatrixTranspose	80.8	1.61×
PageRank	95.6	2.21×
MatrixMultiplication	95.3	1.87×
Im2Col	86.2	4.03×
KMeans	87.2	8.80×
FloydWarshall	95.6	4.93×

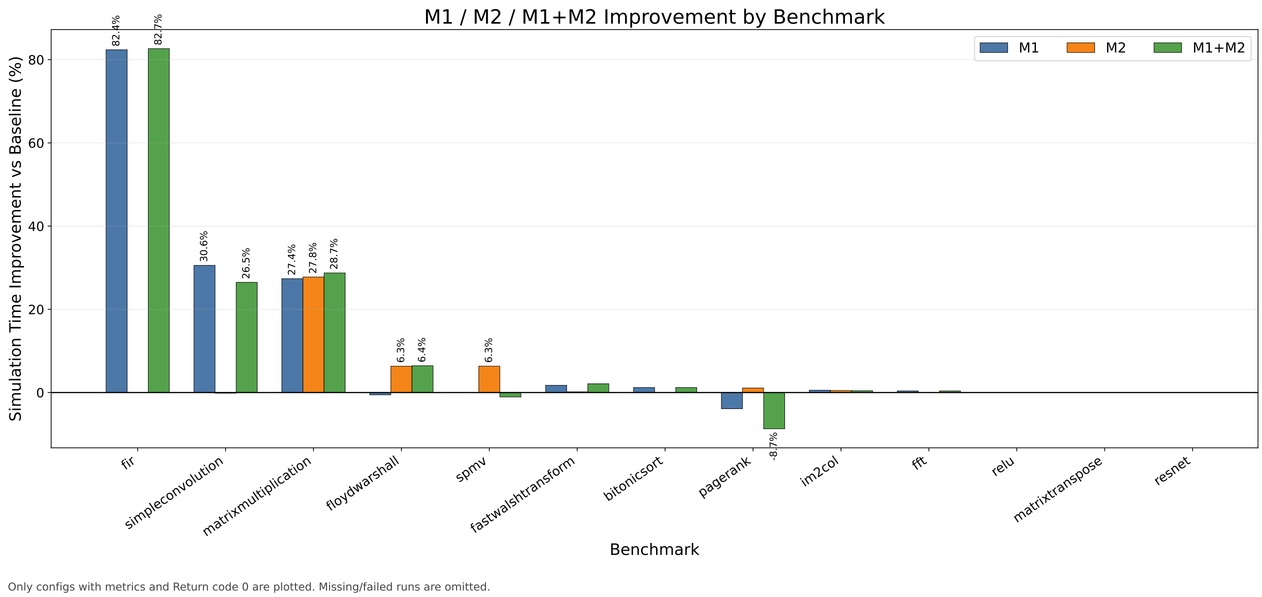

Figure 1: M1/M2 improvement from the June 29 all-mechanisms run.

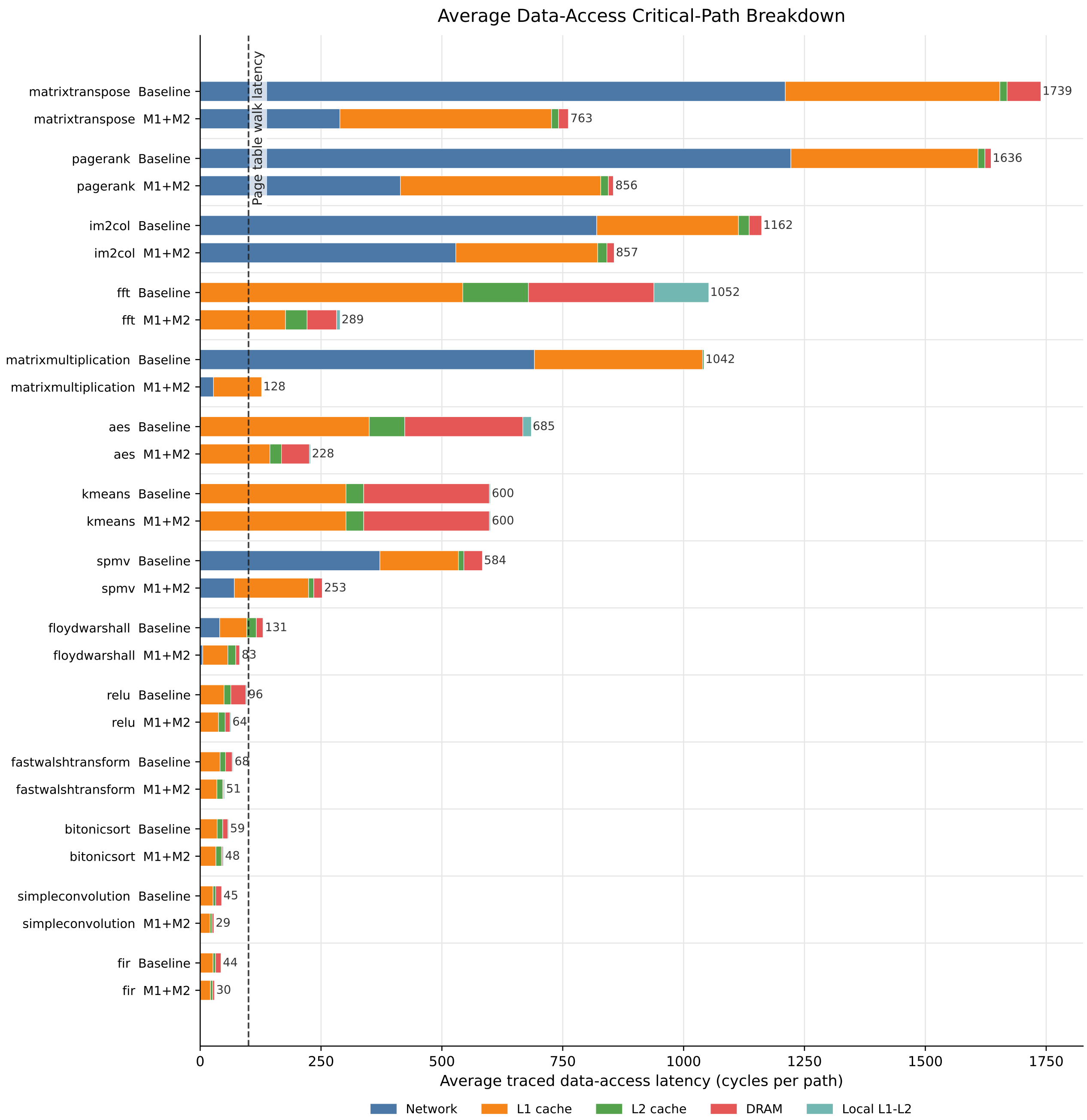

Figure 2: Average data-access critical-path breakdown for baseline and M1+M2. Values are traced latency cycles per memory path, split across network, L1 cache, L2 cache, DRAM, local L1–L2 path, and other components. The dashed vertical line marks a 100-cycle page table walk latency reference.

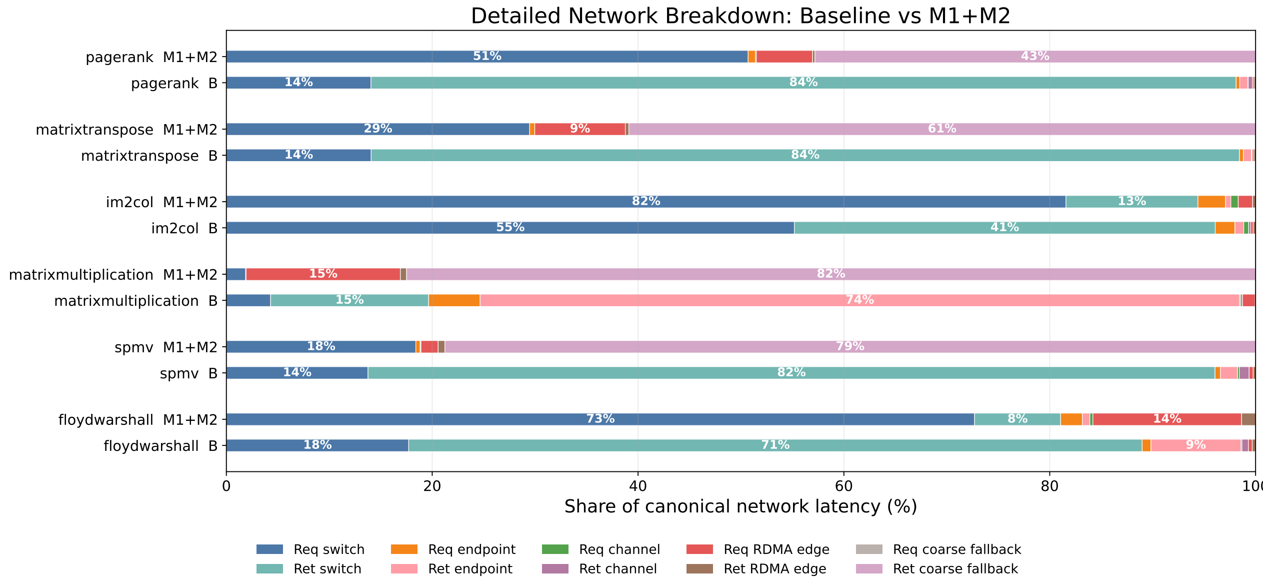

Figure 3: Detailed network breakdown for baseline and M1+M2. Req is the request path from the requester GPU toward the remote owner; Ret is the response path back to the requester. Req/Ret switch are NoC switch input-queue, pipeline, routing, arbitration, and output-wait cycles. Req/Ret endpoint are endpoint queue, injection, assembly, and delivery cycles. Req/Ret channel are link-transfer cycles between endpoints/switches. Req/Ret RDMA edge are cycles at the local/remote RDMA interfaces before entering or after leaving the network. Req/Ret coarse fallback are aggregate RDMA-to-RDMA request/response stages used when fine-grained cross-GPU stages are not available.

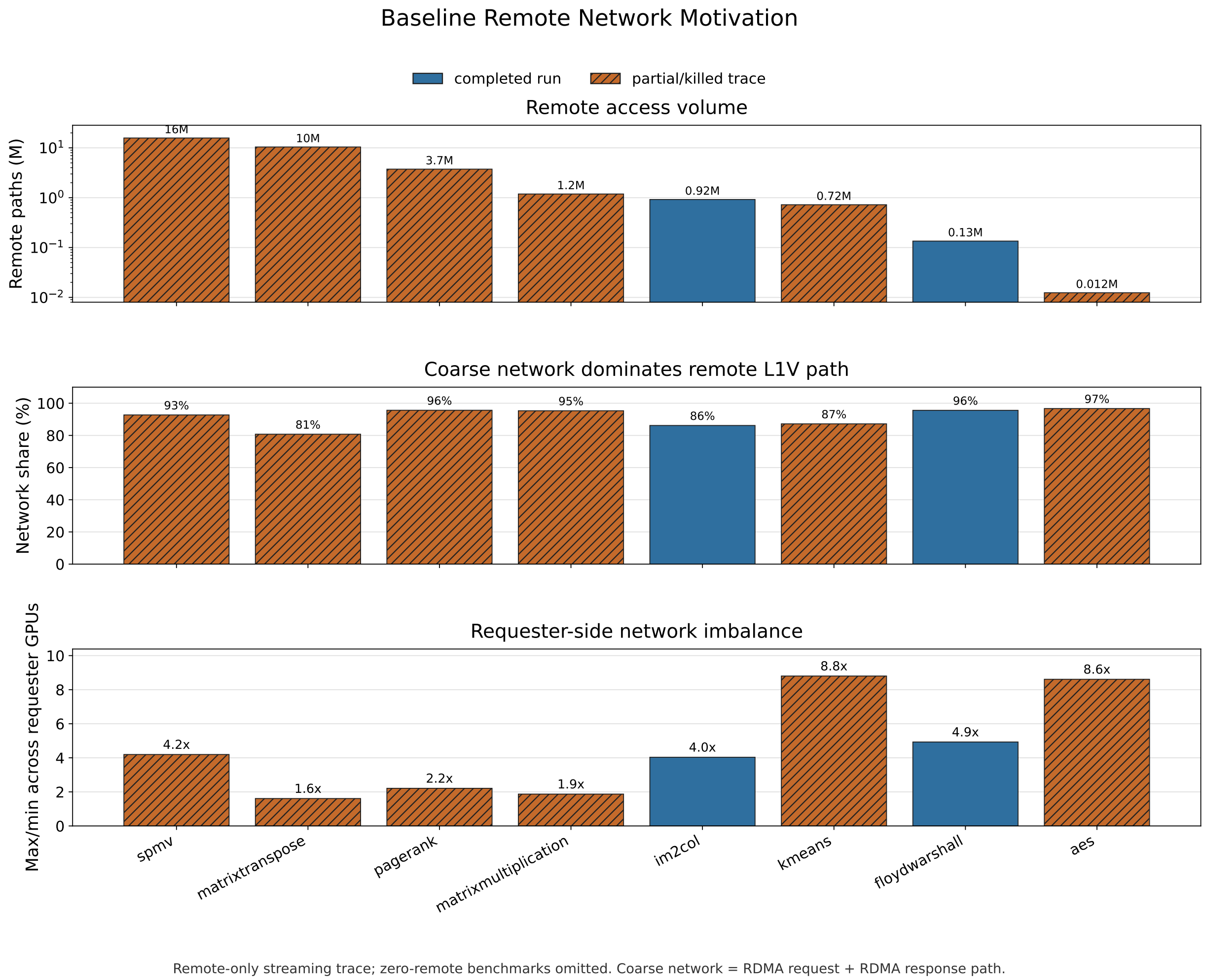

Figure 4: Remote-network motivation from the baseline remote-only trace.